Annex II. Debt Sustainability Analysis

Based on an assessment of public external debt, Moldova's risk of debt distress remains low.[1] *Overall public debt dynamics are sustainable despite a small increase in the debt level in the near term. Private external debt is unusually high for a low-income country. In view of the country's significant vulnerability to exogenous developments and the effects of the banking crisis, fiscal discipline is critical to ensure debt sustainability.*

Background

1. Moldova's total external debt is estimated to reach 100½ percent of GDP by end-2016, compared to 85 percent of GDP in 2014 and 99 percent at end-2015. Private external debt (as a share of GDP), which accounts for about 71 percent of total external debt, is projected to remain stable in 2016, while public and public-guaranteed (PPG) external debt is projected to increase by around 2 percentage points. PPG external debt is held mainly by multilateral and bilateral donors, and is mostly medium and long term.

2. Total PPG debt-to-GDP increased by 9 percentage points to 45 percent at end-2015, and is estimated to be around 46 percent at end–2016. While about two-thirds of PPG debt is external, the share of PPG domestic debt increased rapidly in the past two years due to the issuance of a state guarantee to the NBM to provide emergency liquidity to the banking sector. As the guarantees became callable, the PPG domestic debt structure changed significantly, with about 62 percent of it being the long-term debt securities issued to the NBM. The rest of PPG domestic debt is mainly short term and held by the banking system.

3. External private sector debt to GDP is high, albeit starting to decline.[2] The stock of external private sector debt has decreased in 2015 after a decade of accumulation, reaching US$4.6 billion at end-2015, mostly due to a decrease in long-term bank loans. Because of concerns over the problem banks, the banking system, as a whole, faced difficulties in rolling over debt. As a result, the banks' share of total private external debt dropped to 5¼ percent by end-2015 compared with 9¾ percent by end-2014. The share of medium- and long-term debt in total external private debt was about 52¾ percent at end-2015. About half of non-bank debt is short-term, and consists of trade credits, arrears and other debt liabilities, mostly for the import of natural resources.[3] Similar

[1] This full DSA is prepared jointly by IMF and World Bank staff, in consultation with the Moldovan authorities, using the debt sustainability framework for low-income countries approved by the Boards of both institutions, and in accordance with the new staff guidance note on the application of the joint Bank-Fund Debt Sustainability Framework for Low-Income Countries.

[2] See Table 2, page 49.

[3] Arrears (mainly in the gas sector) increased from US$136.8 million in 2013 to around US$172.3 million in 2014. Other debt liabilities (mainly from other energy sectors) increased from US$537.7 million in 2013 to US$653.7 million in 2014.

to other Central and Eastern European (CEE) countries, private borrowing in Moldova stems mainly from foreign-owned companies borrowing from their parent companies abroad.

4. Moldova has recently become a strong policy performer for the purpose of determining the debt burden thresholds under the Debt Sustainability Framework (DSF). Moldova's rating on the World Bank's Country Policy and Institutional Assessment (CPIA) averaged 3.78 (on a scale of 1 to 6) over the last three years (2013–15), which places the country at the lower end of the strong policy performance category. The corresponding benchmark levels associated with heightened public debt vulnerabilities are presented in the text table.[4] [5]

Debt Thresholds for Strong Policy Performers Under the Debt Sustainability Framework

PPG External Debt Thresholds			
PV of PPG external debt, in percent of:		PPG of external debt service, in percent of:	
GDP + remittances	45	Exports + remittances	20
Export + remittances	160	Revenue	22
Revenue + remittances	300		
Public Debt Benchmark			
PV of total public debt (in percent of GDP)	74		

Source: Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for low-income countries.

Underlying Assumptions

5. The macroeconomic outlook has been revised to reflect changes in near- and medium-term projections compared to the previous DSA assessment. In 2015, the Moldovan economy contracted by 0.5 percent, mainly reflecting negative climatic and external factors together with a decline in remittances, and tight domestic and external financing conditions. As both domestic and external environments remain challenging, the economy is projected to recover modestly in 2016 (with real GDP growth of 2 percent), supported by the rebound in agriculture. Inflation decelerated sharply in early 2016, but has remained above the NBM's target range. This was due to downward adjustments of energy tariffs and a slowdown in food prices. The average inflation is projected to be around 6.8 percent for 2016, taking into account energy tariff adjustments toward the cost-recovery levels take place. The current account deficit is expected to improve in 2016, mainly explained by a sharp reduction in imports due to lower energy prices. Capital flows recorded in the capital and financial accounts are expected to recover gradually over the medium term to the pre-2014 level and allow for an improvement of the reserve position. The fiscal position in 2016 is project to be comparable to the previous assessment, with the cash-based primary deficit of 1.9 percent of GDP

[4] See the staff guidance note on the application of the joint Bank-Fund Debt Sustainability Framework for Low-Income Countries.

[5] Since Moldova's CPIA score is at the lower end of the strong policy performers, any moderate deterioration in macroeconomic management and institutions could push the country into a lower category (e.g., when the next CPIA updates are published). Such a reclassification would reduce the debt threshold levels and may affect the assessment of debt sustainability. For the medium policy performers, the PPG external debt threshold is 36 percent and the public debt benchmark is 56 percent.

(or the augmented primary deficit of 2.2 percent of GDP).[6] For the first seven months of 2016, the stock of arrears has declined by about 0.4 percent of GDP. Without further arrears accumulation, this implies a primary deficit of around 1.5 percent of GDP on an accrual basis.

Box. Macroeconomic Assumptions Behind the DSA

Real GDP is projected to grow 2 percent in 2016, mostly supported by the rebound in agriculture, with remittances remaining subdued. Normalizing the flow of credit in the economy and addressing structural bottlenecks will support potential growth rate at around 4 percent in the medium term.

Inflation is expected to decelerate to 6.8 percent in 2016, reflecting downward adjustments of energy tariffs and a slowdown in food prices. Over the medium term, headline inflation is expected to be around 5 percent, in line with the NBM's target.

Exports of goods and services (in U.S. dollars) are expected to decline by 4.4 percent in 2016, mainly driven by declining unit-value prices of exports. Exports of goods and services, as a ratio of GDP, are projected to reach 40.4 percent in 2016, compared to 43.2 percent in 2015. Over the medium and long term, exports are expected to be supported by structural reforms related to the business environment improvement, better quality and standards of agricultural exports (with supports from other international financial institutions), and cross-border trade liberalization with the EU. As a result, exports as a ratio of GDP are projected to reach 46.3 and 47.7 percent in 2021 and 2036, respectively.

Imports of goods and services (in US dollars) are projected to contract by 6 percent in 2016 due to the decline in global commodity prices and moderate domestic demand (arising from the decline in remittances growth). Imports, as a ratio of GDP, are projected to decline from 73.3 percent in 2015 to 67.3 percent in 2016, and recover to 75.3 percent in the medium term. With favorable external adjustments, imports of goods and services, as a ratio of GDP, are projected to remain at this level in 2036.

Remittances are expected to decline to 19.8 percent of GDP in 2016, mainly due to the recession in Russia and the depreciation of the ruble. In the medium term, remittances are expected to recover to 21.4 percent of GDP, owing to the recovery in the remittance-sourcing countries. In the long term, as the economy develops, the number of migrants is expected to decline due to better domestic employment opportunities, and loss of ties of migrants who remain abroad with the home country. As a result, remittances as a ratio of GDP are projected to gradually decline to 20.5 percent by end-2036.

The current account deficit is projected to narrow in the medium term (5.1 percent of GDP in 2021), following the projected improvement from 3.5 percent in 2016. The current account deficit is expected to remain stable at about 5.2 percent of GDP through 2036. It will be financed by FDI, which is expected to be at about 2.1 percent of GDP in the long run. The reserve level, as a share of GDP, is projected to increase to 31.8 percent in 2021 and stabilize at 34.3 percent of GDP by end-2036.

The overall budget deficit is projected to increase from 2.3 percent of GDP in 2015 to 3.2 percent of GDP in 2016, then decline to around 2.9 percent of GDP by 2019, reflecting the authorities' efforts to ensure fiscal sustainability as specified under the Law on Public Finance and Fiscal Responsibility (FRL) with the use of the escape clause for public capital investment funded by external concessional sources (from 2018 onward). Taking into account on-lending to SOEs as an additional budgetary item, this is equivalent to the augmented budget deficit of about 3.5 percent of GDP in 2016, which gradually declines to 3.1 percent of GDP by 2019. The increased budget deficit in 2016 is due to high interest payments, as well as the reduction of the stock of arrears accumulated in 2015. Over the long run, the primary balance is assumed to be around -1 percent of GDP.

[6] For the program's purpose to appropriately monitor fiscal risks, the augmented primary balance deficit (as presented in the staff report) includes on-lending to SOEs as an additional budgetary item.

Box. Macroeconomic Assumptions Behind the DSA (concluded)

Financing assumptions reflect a shift away from concessional financing. Grant-equivalent financing is projected to increase from 3 percent of GDP expected in 2016 to 3.6 percent of GDP in 2017 before declining to 1.7 percent of GDP in 2021 and further to 1.2 percent of GDP over the long run.

The long-run trend reflects assumptions of declining concessional lending and increasing commercial borrowing (which has a negative weight on the calculation of the grant element of new external borrowing that is shown in Figure 1A), while maintaining the total new external borrowing of 2.5 percent of GDP. Correspondingly, the grant element of new borrowing increases from 29.8 percent in 2016 to 31.5 percent in 2021, and is expected to gradually decline to around 20.8 percent in the long run.

External and Public Debt Sustainability

A. External Debt Sustainability Analysis

6. All external public debt ratios are below the thresholds under the baseline and stress tests scenarios. Under the baseline scenario, the PV of debt-to-GDP and remittances would increase significantly from 25¼ percent in 2015 to 26¾ percent in 2016, and peak at 28½ percent in 2018. This increase in the near term reflects currency depreciation as well as an increase in on-lending agreements with external creditors and an increase in multilateral borrowing as the IMF program is in place.[7] In the long term, the PV of debt-to-GDP and remittances would improve gradually and continuously to around 21½ and 14½ percent in 2026 and 2036, respectively. Under the alternative scenario in which exports grow at their historical average minus one standard deviation in 2017–18, the PV of debt-to-GDP and remittances would peak at 43 percent in 2018, before declining to 30 and 15¼ percent in 2026 and 2036, respectively. Meanwhile, if the key variables remained at their historical averages, the PV of debt to GDP and remittances would continuously increase to around 26 percent in the long run. The PV of debt-to-exports and remittances as well as to revenue would also increase relatively more under the alternative and bound test scenarios, but none of them breach the threshold. This result assumes significant fiscal and external adjustments expected over the medium term, reflecting the authorities' commitment to undertake necessary fiscal and structural reforms.

7. All external public debt ratios of the baseline and stress tests scenarios remain well below the thresholds under the probability approach. The alternative scenario where export growth is set to its historical average minus one standard deviation in 2017–18 results in a near breach of the debt threshold as the peak of the PV of debt-to-GDP in 2018 falls within 10 percent band around the 45 percent threshold. The near breach of the debt threshold under this alternative scenario calls for an analysis which focuses on the evolution of the probability of debt distress over time (incorporating the country's individual CPIA score and average GDP growth rate) in addition to

[7] As discussed in the staff report, both on-lending to the private sector (operationalized through commercial banks) and to SOEs are part of public guarantees and are included in the PPG debt in the DSA.

the evolution of debt burden indicator. The projected probabilities of debt distress associated with various debt burden indicators remain well below their respective thresholds both under the baseline scenario and stress tests.

8. While the external risk rating is determined by the PPG external debt, large private external debt poses additional risks. Private external debt accounted for around 71½ percent of the total external debt in 2015 and it is expected to decline to around 59½ percent in the medium term (text table). Since short-term debt makes up almost half of private external debt, it might be vulnerable to roll-over risks. In addition, about 91½ percent of medium- and long-term private external debts are owned by non-banks private sector, which poses additional risks to official foreign reserves.

9. Furthermore, while exports, remittances, and fiscal revenues are projected to be adequate for the PPG external debt service, the liquidity risk remains. Under the baseline scenario, the PV of PPG external debt service to exports and remittances is expected to increase to 3½ percent in 2016 and peak at 5½ percent in 2019 before declining to around 3¼ percent by end-2036. Similarly, the PV of PPG external debt service to revenues would peak in 2019 and then gradually decline until the end of the projection period. None of the debt service indicators breach the debt service threshold, but some liquidity pressures could emerge. The decline in remittances reflects a downside risk to the external DSA due to the recession in Russia and the depreciation of the ruble.

Breakdown of External Debt, 2013–21

(Millions of U.S. dollars, unless otherwise indicated)

	2013	2014	2015	2016	2017	2018	2019	2020	2021
						Proj.			
Total External Debt	6,682	6,581	6,344	6,622	7,044	7,343	7,495	7,536	7,546
Total PPG	1,758	1,722	1,698	1,887	2,178	2,378	2,479	2,531	2,606
Multilateral	1,572	1,550	1,506	1,646	1,864	2,109	2,274	2,394	2,522
World Bank	603	594	599	675	747	823	896	957	1,012
IMF	772	703	628	597	553	533	485	410	360
EC, EIB, CEDB	128	166	187	244	416	631	804	966	1,119
EBRD	40	56	58	86	101	72	38	6	-26
IFAD	64	62	62	66	68	68	69	70	70
Bilateral	180	171	192	184	157	113	51	-16	-68
Paris Club	168	146	138	131	123	111	97	78	56
Paris Club: ODA	54	50	47	44	40	35	30	24	19
Paris Club: non-ODA	99	78	73	69	64	57	48	37	23
Other Official Bilateral	12	25	54	53	34	2	-46	-94	-125
Commercial	5	1	0	0	0	0	0	0	0
Publicly guarenteed assumed debt/private debt	2	0	0	56	157	156	153	152	152
Total Private	4,924	4,859	4,645	4,736	4,865	4,965	5,016	5,006	4,941
Loans	2,801	2,653	2,510	2,540	2,564	2,560	2,500	2,381	2,204
Short terms	111	56	61	66	71	86	104	123	131
Banks	43	37	37	51	55	66	80	95	101
Other private sectors	69	19	24	15	16	20	24	28	30
Medium and Long terms	2,689	2,597	2,449	2,475	2,493	2,474	2,396	2,258	2,072
Banks	433	301	205	383	386	383	371	349	321
Other private sectors	2,256	2,296	2,244	2,092	2,108	2,091	2,025	1,909	1,752
Other short term	2,123	2,206	2,135	2,195	2,301	2,405	2,517	2,624	2,737
Currency and deposits	371	234	101	108	114	121	128	135	143
Trade credits	1,067	1,137	1,147	1,200	1,300	1,397	1,502	1,602	1,707
Other debt liabilities	686	836	887	887	887	887	887	887	887

Source: Country authorities; and staff estimates and projections

B. Public Sector Debt Sustainability Analysis

10. Under the baseline scenario, the PPG debt-to-GDP ratio is projected to increase by about 0.9 to 1 percentage point of GDP annually during 2016–18—before declining over the medium and long term, reflecting sustainable public debt dynamics. The PPG debt-to-GDP ratio would increase from 45 percent in 2015 to about 46, 47 and 48 percent of GDP in 2016, 2017, and 2018, respectively. The domestic PPG debt-to-GDP ratio is projected to decline, while the increase in the external PPG debt-to-GDP ratio is mainly driven by the currency depreciation as well as an increase in on-lending agreements and an increase in multilateral borrowing (given that the IMF program is in place). With the macroeconomic assumptions outlined above, the recommended (augmented) primary budget deficit path is projected to move towards the path which is below the level that would stabilize the debt-to-GDP ratio. As a result, Moldova's total PPG debt is expected to decline gradually and continuously from its peak in 2018 to 47 percent in 2021 and 32 percent by end-2036. Other sustainability indicators confirm a similar long-term trend under the baseline scenario. PV of debt-to-GDP is projected to increase slightly to 47½ percent in 2016, and then gradually improve to 46 percent and 30¾ percent in 2021 and 2036, respectively. These PV of debt projections are below the benchmark level of 74 percent associated with heightened public debt vulnerabilities for the strong policy performance category. Similarly, the PV of debt-to-revenue and grants ratio is projected to initially increase to 135.3 percent of GDP in 2016, and decline continuously to 130½ percent and 86¾ percent by 2021 and end-2036 respectively.

11. While the inclusion of domestic debt does not alter the assessment of Moldova's overall risk of debt distress, it emphasizes the importance of prudent fiscal and borrowing policies for the preservation of the low risk rating. Under all standard alternative scenarios presented in the DSA, the PV of public debt-to-GDP is projected to be well below the benchmark level of 74 percent. However, some alternative scenarios would raise the public debt level faster than others. For example, a 30 percent real depreciation in 2017 would have the strongest impact in the short run with the PV of debt-to-GDP ratio reaching 60 percent in 2017. Similarly, a 10 percent of GDP increase in other debt creating flows in 2017 would cause the PV of debt-to-GDP to peak at 55 percent in 2017–18. For example, such risks could stem from contingencies for the rest of the banking system and for the accumulated energy tariff debt.

C. The Authorities' View

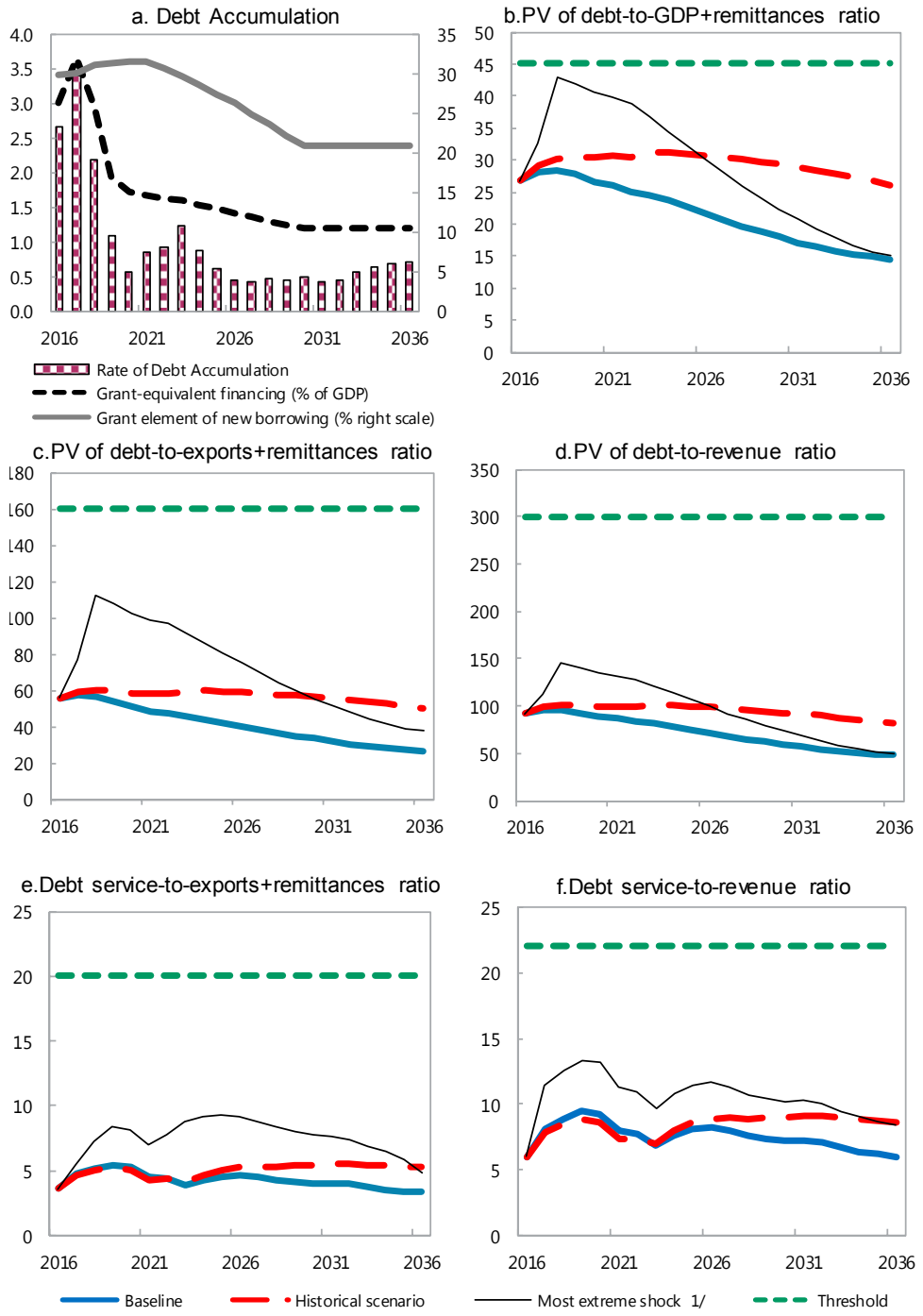
12. The authorities concurred with the staff assessment, and noted the importance of fiscal and external sustainability. The authorities acknowledged that maintaining prudent fiscal policies over the medium term would help protect against potential fiscal risks and liabilities, and hence would strengthen debt sustainability. They also agreed that it is necessary to strengthen the framework to monitor and contain fiscal risks. In addition, as guided by the FRL, the authorities stressed the need for flexibility to be able to finance capital investment in particular when projects are financed by concessional lending. These public investment projects are essential to making progress in poverty reduction, filling infrastructure gaps, and institution strengthening.

D. Conclusion

13. The DSA indicates that Moldova's risk of debt distress remains low, in line with the previous assessment at end-2015. All external indicators for public debt remain well below the debt thresholds under the baseline, standard bound tests, and alternative scenarios as well as under the probability approach. However, significant private external debt poses roll-over risks to debt sustainability. Likewise, while being more sensitive to exchange rate depreciation and a sudden increase in other debt creating flows, Moldova's overall public debt dynamics are projected to remain on a sustainable path under the baseline scenario and alterative scenarios.

14. Pursuing prudent fiscal policy and advancing structural reforms remain necessary to ensure debt sustainability. Due to the country's sensitivity to exogenous developments and the banking crisis, debt sustainability critically depends on sound macroeconomic management and continuing progress on institutional and structural issues that would help unlock the economy's growth potential and reduce its vulnerability to shocks. Furthermore, the limited development of the domestic debt market poses financing risks, especially considering the country's development needs and significant dependence on foreign assistance in the form of grants and concessional loans. Efforts to lengthen the average maturity of domestic debt and deepen the secondary market would help reduce the PPG domestic debt roll-over and interest rate risks.

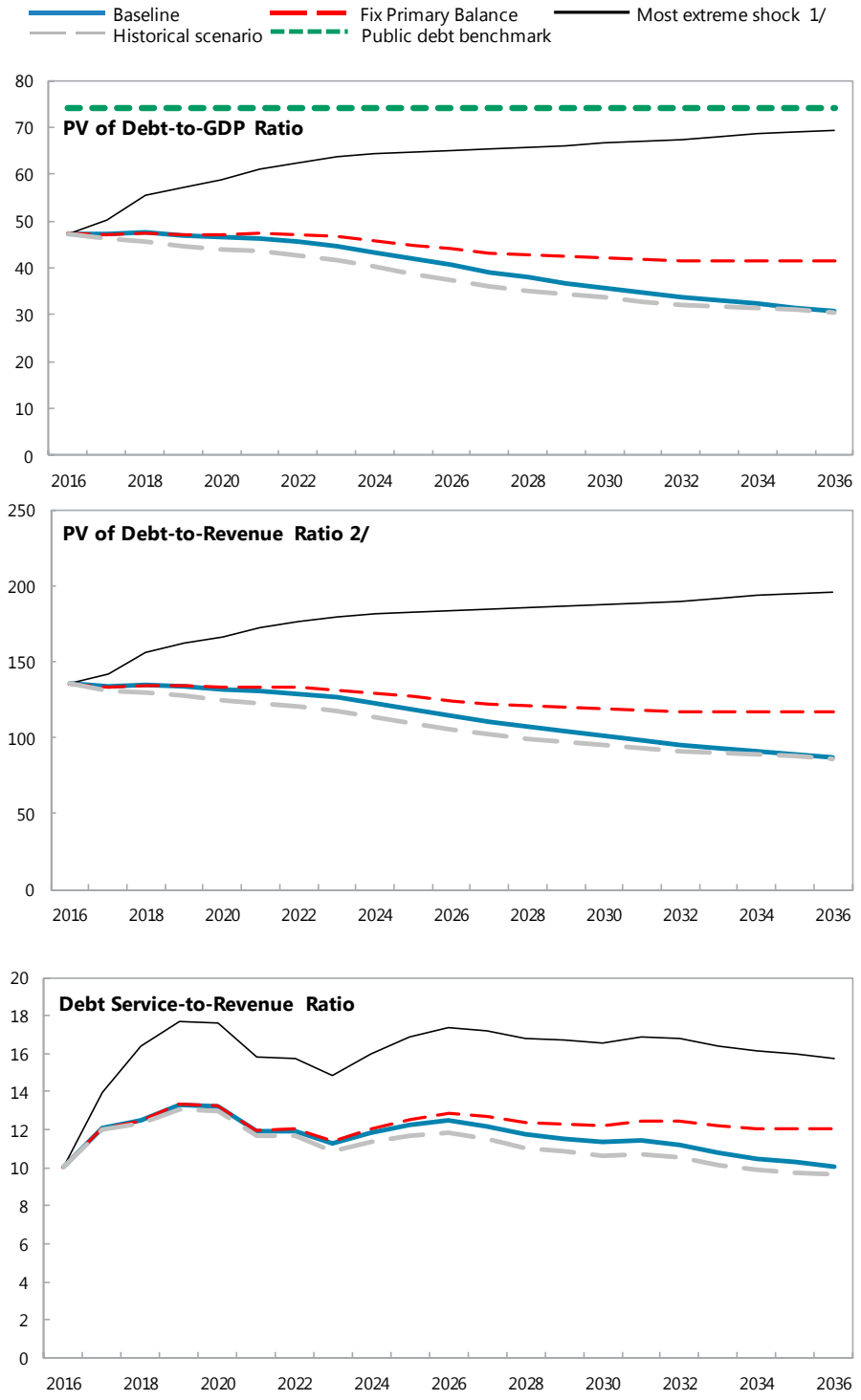
Figure 1A. Moldova: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2016–36 1/



Sources: Country authorities; and staff estimates and projections.

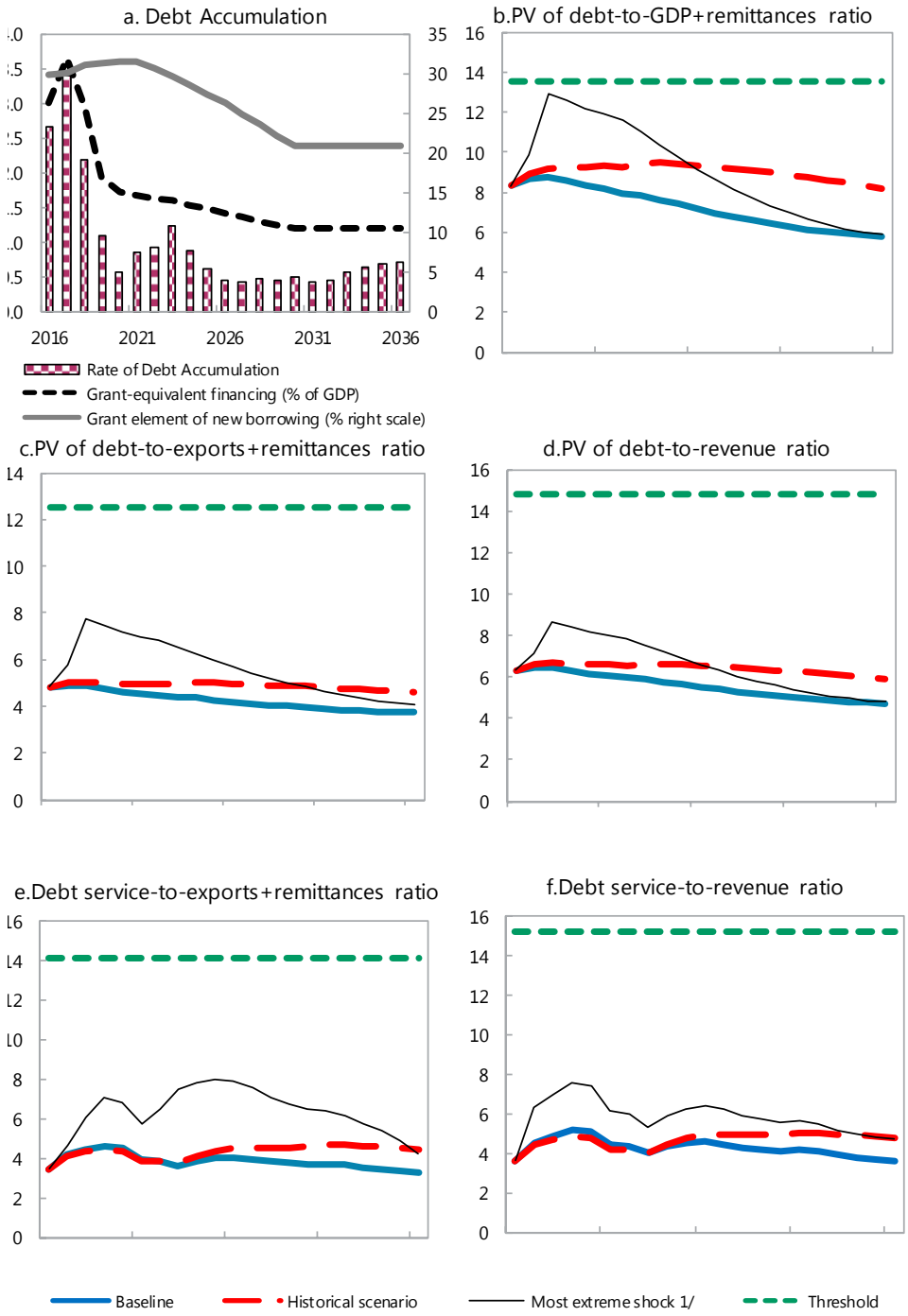
1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to an Exports shock; in c. to an Exports shock; in d. to an Exports shock; in e. to an Exports shock and in figure f. to a one-time depreciation shock.

Figure 2A. Moldova: Indicators of Public Debt Under Alternative Scenarios, 2016–36 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2026.
2/ Revenues are defined inclusive of grants.

Figure 3A. Moldova: Probability of Debt Distress of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2016–36 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to an Exports shock; in c. to an Exports shock; in d. to an Exports shock; in e. to an Exports shock and in figure f. to a one-time depreciation shock.

Table 1A. Moldova: External Debt Sustainability Framework, Baseline Scenario, 2013–36 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2013	2014	2015			2016	2017	2018	2019	2020	2021	2016-2021 Average	2026	2036	2022-2036 Average
External debt (nominal) 1/	84.5	84.9	98.9			100.4	101.2	100.3	96.7	91.6	87.1		74.4	60.1	
of which: public and publicly guaranteed (PPG)	22.8	24.0	27.3			29.1	31.9	33.0	32.5	31.2	30.5		25.3	17.9	
Change in external debt	1.8	0.4	14.0			1.5	0.9	-1.0	-3.6	-5.0	-4.5		-2.4	-1.7	
Identified net debt-creating flows	-4.2	3.2	20.5			-0.5	0.1	0.1	-0.2	-0.3	-0.2		0.4	0.9	
Non-interest current account deficit	5.1	4.4	2.9	8.9	4.4	2.3	3.4	3.5	3.1	3.1	3.0		3.2	4.5	3.5
Deficit in balance of goods and services	37.4	36.7	30.1			26.9	27.7	28.4	28.7	28.9	29.0		28.7	28.0	
Exports	43.3	41.5	43.2			40.4	41.7	43.0	44.2	45.4	46.3		46.8	47.7	
Imports	80.7	78.2	73.3			67.3	69.3	71.3	72.8	74.2	75.3		75.5	75.8	
Net current transfers (negative = inflow)	-20.7	-19.8	-18.0	-21.7	2.9	-16.4	-16.3	-16.2	-16.0	-16.1	-16.0		-15.8	-15.3	-15.7
of which: official	-1.9	-2.6	-2.3			-1.5	-1.2	-1.0	-0.7	-0.7	-0.5		-0.7	-0.7	
Other current account flows (negative = net inflow)	-11.6	-12.5	-9.2			-8.2	-8.0	-8.7	-9.5	-9.6	-9.9		-9.6	-8.2	
Net FDI (negative = inflow)	-2.7	-2.0	-2.5	-5.2	3.8	-2.1	-2.0	-2.0	-1.9	-2.0	-2.0		-1.9	-2.1	-2.0
Endogenous debt dynamics 2/	-6.6	0.8	20.2			-0.7	-1.2	-1.5	-1.4	-1.4	-1.3		-0.9	-1.5	
Contribution from nominal interest rate	0.7	0.8	0.8			1.3	1.6	1.9	2.1	2.1	2.0		1.9	0.7	
Contribution from real GDP growth	-7.1	-4.1	0.5			-1.9	-2.9	-3.4	-3.5	-3.5	-3.3		-2.7	-2.2	
Contribution from price and exchange rate changes	-0.2	4.1	19.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	6.0	-2.9	-6.5			1.9	0.7	-1.0	-3.4	-4.8	-4.3		-2.8	-2.6	
of which: exceptional financing	-1.5	-4.2	-1.4			-0.9	-1.2	-0.9	-0.6	-0.6	-0.6		-0.6	-0.6	
PV of external debt 4/	...	...	100.7			101.9	101.7	100.1	96.2	91.1	86.6		73.9	58.8	
In percent of exports	...	...	233.0			252.4	244.0	232.9	217.9	200.9	186.8		158.0	123.2	
PV of PPG external debt	...	...	29.2			30.7	32.4	32.8	32.0	30.7	30.0		24.8	16.7	
In percent of exports	...	...	67.5			76.0	77.7	76.3	72.5	67.7	64.7		53.0	34.9	
In percent of government revenues	...	...	85.4			92.2	96.4	96.2	93.0	88.6	86.2		71.4	47.9	
Debt service-to-exports ratio (in percent)	17.3	15.0	12.8			26.1	22.1	28.1	28.3	26.2	24.0		18.8	8.0	
PPG debt service-to-exports ratio (in percent)	2.6	2.6	2.4			4.9	6.5	7.0	7.4	7.1	6.0		6.1	4.4	
PPG debt service-to-revenue ratio (in percent)	3.3	3.2	3.1			6.0	8.1	8.9	9.4	9.3	8.0		8.3	6.0	
Total gross financing need (Millions of U.S. dollars)	2623.4	2924.9	2641.3			2912.1	3003.5	3375.1	3563.3	3694.7	3809.1		4638.4	8371.4	
Non-interest current account deficit that stabilizes debt ratio	3.3	4.1	-11.2			0.8	2.5	4.5	6.7	8.1	7.6		5.6	6.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	9.4	4.8	-0.5	3.7	4.8	2.0	3.0	3.6	3.7	3.8	3.8	3.3	3.8	3.8	3.8
GDP deflator in US dollar terms (change in percent)	0.2	-4.6	-18.3	5.2	14.0	0.4	2.5	1.5	2.0	2.2	1.5	1.7	2.9	2.9	2.9
Effective interest rate (percent) 5/	0.9	0.9	0.7	1.3	0.3	1.3	1.7	2.0	2.2	2.3	2.4	2.0	2.6	1.2	2.3
Growth of exports of G&S (US dollar terms, in percent)	10.3	-4.2	-15.3	8.0	19.2	-4.4	9.0	8.4	8.8	9.0	7.6	6.4	7.1	7.1	7.0
Growth of imports of G&S (US dollar terms, in percent)	6.3	-3.0	-23.8	8.2	23.1	-6.0	8.8	8.1	8.1	8.1	6.9	5.7	6.9	6.9	6.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	29.8	30.0	31.0	31.2	31.5	31.5	30.8	26.2	20.8	23.9
Government revenues (excluding grants, in percent of GDP)	34.8	34.4	34.2			33.3	33.6	34.1	34.4	34.7	34.8		34.8	34.8	34.8
Aid flows (in Millions of US dollars) 7/	186.4	332.0	147.7			109.8	120.8	96.1	53.0	55.9	58.9		81.1	157.3	
of which: Grants	154.6	278.8	101.3			109.8	120.8	96.1	53.0	55.9	58.9		81.1	157.3	
of which: Concessional loans	31.8	53.2	46.4			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.0	3.6	2.9	1.9	1.7	1.7		1.4	1.2	1.3
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			48.6	44.7	44.8	41.2	43.0	43.7		40.5	37.6	39.2
Memorandum items:															
Nominal GDP (Millions of US dollars)	7984.9	7983.2	6492.8			6650.6	7019.4	7378.3	7810.9	8286.0	8730.2		12008.4	23299.8	
Nominal dollar GDP growth	9.6	0.0	-18.7			2.4	5.5	5.1	5.9	6.1	5.4	5.1	6.9	6.9	6.8
PV of PPG external debt (in Millions of US dollars)			1814.8			1986.9	2211.7	2364.7	2444.4	2488.9	2558.1		2949.6	3841.5	
(PVt-PVt-1)/GDPt-1 (in percent)						2.6	3.4	2.2	1.1	0.6	0.8	1.8	0.4	0.7	0.6
Gross workers' remittances (Millions of US dollars)	1501.4	1380.4	1021.7			989.3	1058.7	1122.1	1196.0	1276.7	1352.1		1809.5	3402.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	25.2			26.7	28.1	28.5	27.8	26.6	25.9		21.6	14.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	49.5			55.6	57.1	56.3	53.8	50.6	48.5		40.1	26.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.8			3.6	4.8	5.2	5.5	5.3	4.5		4.6	3.3	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g))/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes. The high levels of residuals in the external DSA is mostly explained by exchange rate valuation.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1B. Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016–36

(In percent)

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of debt-to-GDP+remittances ratio								
Baseline	27	28	28	28	27	26	22	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	27	29	30	30	30	31	31	26
A2. New public sector loans on less favorable terms in 2016-2036 2	27	28	30	29	29	29	27	24
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	27	28	30	29	28	27	23	15
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	27	33	43	42	41	40	30	15
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	27	30	34	33	31	31	26	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	27	26	25	24	23	23	20	14
B5. Combination of B1-B4 using one-half standard deviation shocks	27	28	33	33	32	31	25	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	27	37	37	36	35	34	29	19
PV of debt-to-exports+remittances ratio								
Baseline	56	57	56	54	51	48	40	27
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	56	59	60	60	59	58	59	50
A2. New public sector loans on less favorable terms in 2016-2036 2	56	58	59	57	55	54	50	44
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	56	55	55	53	49	47	40	26
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	56	77	112	108	102	99	75	37
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	56	55	55	53	49	47	40	26
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	56	51	48	47	44	43	37	26
B5. Combination of B1-B4 using one-half standard deviation shocks	56	56	66	66	62	60	49	30
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	56	55	55	53	49	47	40	26
PV of debt-to-revenue ratio								
Baseline	92	96	96	93	89	86	71	48
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	92	99	101	101	100	100	99	81
A2. New public sector loans on less favorable terms in 2016-2036 2	92	97	100	99	96	95	89	80
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	92	98	103	99	94	92	77	52
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	92	112	145	140	135	132	99	50
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	92	105	118	114	108	105	88	59
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	92	89	85	82	78	76	65	47
B5. Combination of B1-B4 using one-half standard deviation shocks	92	100	119	115	110	107	87	54
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	92	133	133	128	122	119	100	67

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

Table 1B. Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016–36 (concluded)

(In percent)

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
Debt service-to-exports+remittances ratio								
Baseline	4	5	5	5	5	4	5	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	4	5	5	5	5	4	5	5
A2. New public sector loans on less favorable terms in 2016-2036 2	4	5	5	6	6	5	5	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	4	5	5	5	5	4	5	3
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	4	6	7	8	8	7	9	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	4	5	5	5	5	4	5	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	4	5	5	5	5	4	4	3
B5. Combination of B1-B4 using one-half standard deviation shocks	4	5	6	6	6	5	6	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	4	5	5	5	5	4	5	3
Debt service-to-revenue ratio								
Baseline	6	8	9	9	9	8	8	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	6	8	8	9	9	7	9	9
A2. New public sector loans on less favorable terms in 2016-2036 2	6	8	9	10	10	9	9	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	6	8	10	10	10	9	9	7
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	6	8	9	11	11	9	12	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	6	9	11	12	12	10	10	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	6	8	9	9	9	8	8	6
B5. Combination of B1-B4 using one-half standard deviation shocks	6	9	10	11	11	9	10	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	6	11	13	13	13	11	12	8
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	21	21	21	21	21	21	21	21

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2A. Moldova: Public Sector Debt Sustainability Framework, Baseline Scenario, 2013–36

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2013	2014	2015	Average 5/	Standard Deviation 5/	2016	2017	2018	2019	2020	2021	2016-21 Average	2026	2036	2022-36 Average
Public sector debt 1/	29.6	36.0	45.0			45.7	46.8	47.7	47.3	46.9	46.8		41.1	32.0	
of which: foreign-currency denominated	22.8	24.0	27.3			29.1	31.9	33.0	32.5	31.2	30.5		25.3	17.9	
Change in public sector debt	-1.3	6.5	8.9			0.7	1.1	0.9	-0.4	-0.4	-0.1		-1.5	-0.6	
Identified debt-creating flows	-0.5	8.1	10.8			0.8	1.0	0.4	-0.1	-0.1	0.0		-0.5	-0.3	
(Augmented) Primary deficit	1.3	1.3	1.4	1.2	1.7	2.2	2.3	2.0	1.7	1.7	1.6	1.9	1.3	0.8	1.1
Revenue and grants	36.7	37.9	35.7			34.9	35.3	35.4	35.1	35.4	35.4		35.4	35.4	
of which: grants	1.9	3.5	1.6			1.7	1.7	1.3	0.7	0.7	0.7		0.7	0.7	
(Augmented) Primary (noninterest) expenditure	38.1	39.2	37.2			37.1	37.7	37.4	36.8	37.0	37.1		36.7	36.2	
Automatic debt dynamics	-1.5	1.5	3.6			-1.1	-1.2	-1.5	-1.6	-1.7	-1.6		-1.8	-1.1	
Contribution from interest rate/growth differential	-2.7	-1.5	-0.2			-0.8	-0.9	-1.5	-1.7	-1.7	-1.7		-1.5	-1.0	
of which: contribution from average real interest rate	-0.1	-0.2	-0.4			0.1	0.5	0.1	0.1	0.0	0.0		0.0	0.2	
of which: contribution from real GDP growth	-2.7	-1.4	0.2			-0.9	-1.3	-1.6	-1.7	-1.7	-1.7		-1.6	-1.2	
Contribution from real exchange rate depreciation	1.2	3.1	3.9			-0.3	-0.3	0.0	0.0	0.0	0.2		...	...	
Other identified debt-creating flows	-0.3	5.2	5.7			-0.2	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
Privatization receipts (negative)	-0.3	-0.4	0.0			-0.2	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	5.6	5.8			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-0.8	-1.7	-1.8			0.0	0.0	0.5	-0.3	-0.2	-0.1		-1.0	-0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	46.8			47.3	47.3	47.5	46.9	46.4	46.2		40.5	30.7	
of which: foreign-currency denominated	...	...	29.2			30.7	32.4	32.8	32.0	30.7	30.0		24.8	16.7	
of which: external	...	...	29.2			30.7	32.4	32.8	32.0	30.7	30.0		24.8	16.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	9.2	9.2	9.3			11.5	12.4	11.4	11.9	12.7	13.7		16.3	16.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	131.0			135.3	133.8	134.4	133.6	131.3	130.5		114.4	86.7	
PV of public sector debt-to-revenue ratio (in percent)	...	...	137.0			142.0	140.7	139.5	136.2	133.8	133.1		116.7	88.4	
of which: external 3/	...	...	85.4			92.2	96.4	96.2	93.0	88.6	86.2		71.4	47.9	
Debt service-to-revenue and grants ratio (in percent) 4/	4.9	5.3	6.1			10.0	12.1	12.5	13.3	13.2	11.9		12.5	10.0	
Debt service-to-revenue ratio (in percent) 4/	5.2	5.8	6.4			10.5	12.7	13.0	13.6	13.5	12.2		12.7	10.2	
Primary deficit that stabilizes the debt-to-GDP ratio	2.7	-5.1	-7.5			1.4	1.3	1.1	2.1	2.0	1.8		2.8	1.4	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	9.4	4.8	-0.5	3.7	4.8	2.0	3.0	3.6	3.7	3.8	3.8	3.3	3.8	3.8	3.8
Average nominal interest rate on forex debt (in percent)	0.9	0.9	0.8	1.4	0.5	1.1	1.6	1.7	1.8	1.7	1.8	1.6	1.9	2.8	2.3
Average real interest rate on domestic debt (in percent)	1.0	-0.5	-3.1	0.3	6.0	0.1	2.4	1.1	1.2	1.1	0.5	1.1	0.4	0.5	0.5
Real exchange rate depreciation (in percent, + indicates depreciation)	5.6	14.3	16.1	-1.4	13.8	-1.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.1	6.4	9.6	8.7	4.1	6.5	4.5	5.0	5.0	5.0	5.0	5.2	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	5.5	8.0	-5.7	0.8	3.7	1.8	4.6	2.8	2.0	4.5	3.9	3.3	3.2	3.7	3.6
Grant element of new external borrowing (in percent)	...	...	...	...	...	29.8	30.0	31.0	31.2	31.5	31.5	30.8	26.2	20.8	...

Sources: Country authorities; and staff estimates and projections.

1/ Public sector debt covers gross debt of the general government. Debt of state-owned enterprises is not included unless it is explicitly guaranteed by the government.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2B. Moldova: Sensitivity Analysis for Key Indicators of Public Debt 2016–36

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of Debt-to-GDP Ratio								
Baseline	47	47	48	47	46	46	41	31
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	47	46	46	45	44	44	37	31
A2. Primary balance is unchanged from 2016	47	47	47	47	47	47	44	41
A3. Permanently lower GDP growth 1/	47	48	49	50	51	53	60	92
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	47	50	55	57	59	61	65	69
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	47	48	49	48	48	47	41	31
B3. Combination of B1-B2 using one half standard deviation shocks	47	48	49	50	50	51	50	47
B4. One-time 30 percent real depreciation in 2017	47	60	59	58	57	56	49	39
B5. 10 percent of GDP increase in other debt-creating flows in 2017	47	55	55	54	54	54	47	35
PV of Debt-to-Revenue Ratio 2/								
Baseline	135	134	134	134	131	131	114	87
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	135	131	129	127	124	123	105	86
A2. Primary balance is unchanged from 2016	135	133	134	135	133	134	124	117
A3. Permanently lower GDP growth 1/	135	136	139	142	144	149	168	258
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	135	142	156	162	166	172	183	195
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	135	135	138	137	134	134	117	88
B3. Combination of B1-B2 using one half standard deviation shocks	135	135	139	141	142	144	141	132
B4. One-time 30 percent real depreciation in 2017	135	169	167	165	161	159	139	110
B5. 10 percent of GDP increase in other debt-creating flows in 2017	135	156	156	155	152	151	132	98
Debt Service-to-Revenue Ratio 2/								
Baseline	10	12	12	13	13	12	12	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	10	12	12	13	13	12	12	10
A2. Primary balance is unchanged from 2016	10	12	12	13	13	12	13	12
A3. Permanently lower GDP growth 1/	10	12	13	14	14	13	15	22
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	10	13	14	15	15	14	17	19
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	10	12	13	13	13	12	13	10
B3. Combination of B1-B2 using one half standard deviation shocks	10	12	13	14	14	13	14	14
B4. One-time 30 percent real depreciation in 2017	10	14	16	18	18	16	17	16
B5. 10 percent of GDP increase in other debt-creating flows in 2017	10	12	13	14	14	13	14	11

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.